Volume & Pressure Changes in the Lungs

Exam code: 0610 & 0970

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Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
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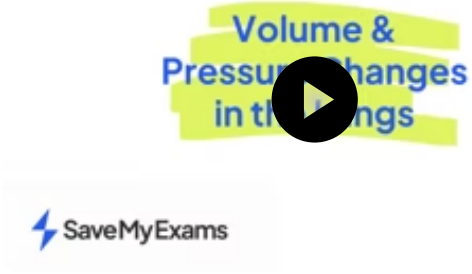

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Volume & Pressure Changes in the Lungs: Extended

Extended Tier Only

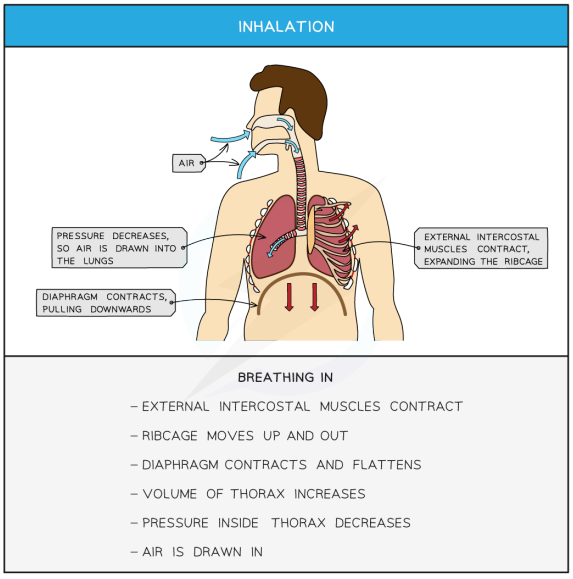
- The **diaphragm** is a thin sheet of muscle that separates the chest cavity from the abdomen; it is ultimately responsible for **controlling ventilation** in the lungs
  - When the diaphragm **contracts** it flattens and this **increases the volume** of the chest cavity (thorax), which consequently leads to a **decrease in air pressure** inside the lungs relative to outside the body, **drawing air in**.
  - When the diaphragm **relaxes** it moves upwards back into its domed shape and this **decreases the volume** of the chest cavity (thorax), which consequently leads to an **increase in air pressure** inside the lungs relative to outside the body, **forcing air out**
- The external and internal intercostal muscles work as **antagonistic** pairs (meaning they work in different directions to each other)
- During **inhalation** the **external set of intercostal muscles contract** to pull the ribs **up and out:**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

relax so the ribs **drop down and in:**
    - **This decreases the volume of the chest cavity (thorax) increasing air pressure, forcing air out**

- When we need to increase the rate of gas exchange (for example during strenuous activity) the internal intercostal muscles will also work to **pull the ribs down and in** to decrease the volume of the thorax more, forcing air out more forcefully and quickly – this is called **forced exhalation**
    - There is actually a greater need to rid the body of increased levels of carbon dioxide produced during strenuous activity!

- This allows a **greater volume of gases to be exchanged**



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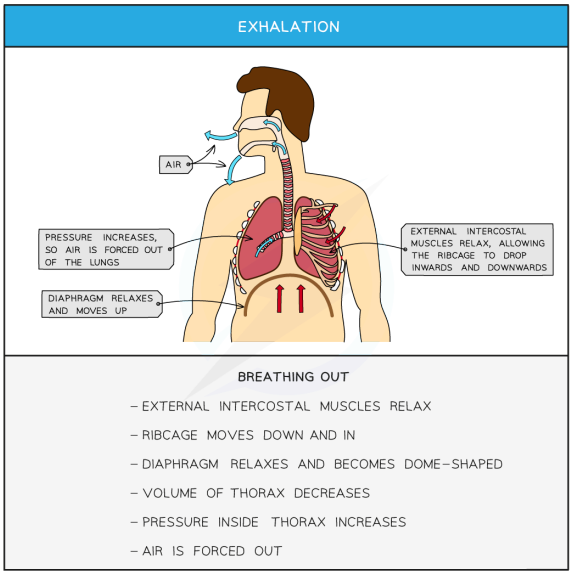

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Examiner Tips and Tricks

You may see the terms **inhalation OR inspiration** (breathing in), and **exhalation OR expiration** (breathing out). Both sets of terms mean exactly the same thing, so don't let them confuse you!This sequence of events is a common exam question and you should be able to **explain in detail** what is happening to the external and internal intercostal muscles, the rib cage, the diaphragm, the volume and the pressure-volume of the lungs when breathing in and out.Remember, if you learn one, the other is almost exactly the opposite.

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